

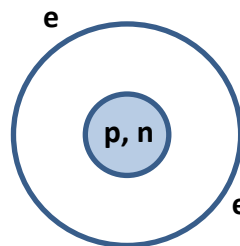


Study Prep

Available on Pearson+

CONCEPT: ELECTRIC CHARGE

- Atoms are made of protons, neutrons and electrons.



- ELECTRIC CHARGE is a property of matter, similar to MASS:

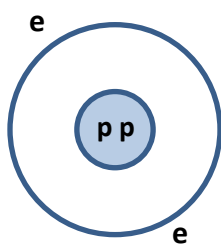
MASS (m)	ELECTRIC CHARGE (Q)
- Mass → Gravitational Force - More Mass → More Gravity - Mass → ONLY _____	- Electric Charge → Electric Force - More Charge → More Electric Force - Charge → _____ and _____

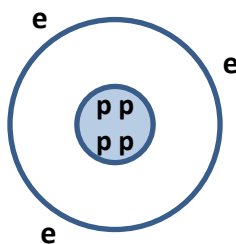
- ELEMENTARY charge → $e = \text{_____ C}$ (_____)

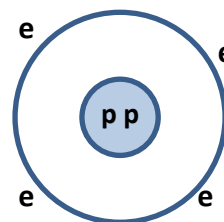
- Charge of protons = _____

- Charge of electrons = _____

- The CHARGE of an object is the quantity of _____ of protons and electrons in it:







- Notice these charges are in WHOLE MULTIPLES of e .

→ $Q = (\# \text{prot} - \# \text{elec}) \times e$

- MOST materials are NEUTRAL → #Protons _____ #Electrons → $Q_{\text{net}} = \text{_____}$



PRACTICE: CHARGE OF ATOM

What is the charge of an atom with 16 protons and 7 electrons?

EXAMPLE: NUMBER OF ELECTRONS

How many electrons make up $-1.5 \times 10^{-5} \text{ C}$?



EXAMPLE: ELECTRONS IN WATER

Water has a density of 1 kg/L, a molecular weight of 18 g/mol, and 10 electrons per molecule.

- a) How many electrons does 2L of water have?
- b) What charge do these electrons represent?

PRACTICE: ADDING ELECTRONS

How many electrons do you have to add to decrease the charge of an object by $16\mu\text{C}$?

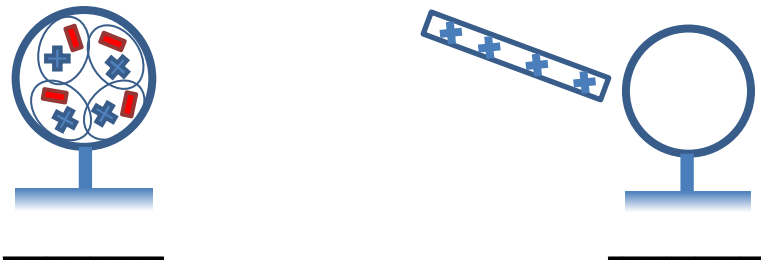
CONCEPT: CHARGING OBJECTS

- Electricity: The _____ of electrons & electric charge.
- Materials come in two types: _____ and _____
 - Conductors (**ALLOW** | **DON'T ALLOW**) electrons/charges to move (e.g. metals)
 - Insulators (**ALLOW** | **DON'T ALLOW**) electrons/charges to move (e.g. plastics, rubber)
- Rubbing objects together strips electrons from one and gives to the other
 - Fur & plastic rod = rod has — charge - Fur & glass rod = rod has + charge

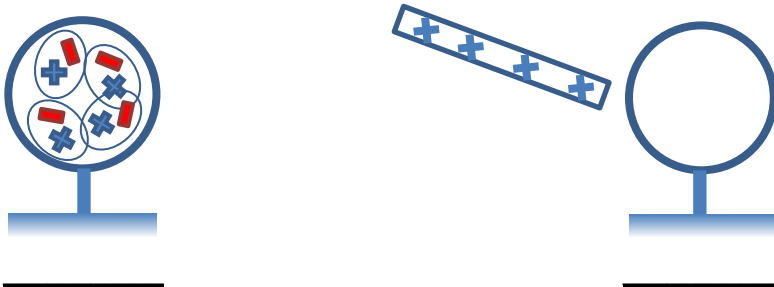
• Like charges (**ATTRACT** | **REPEL**) and unlike charges (**ATTRACT** | **REPEL**).

- POLARIZATION = separation of charges → NO NET CHARGE

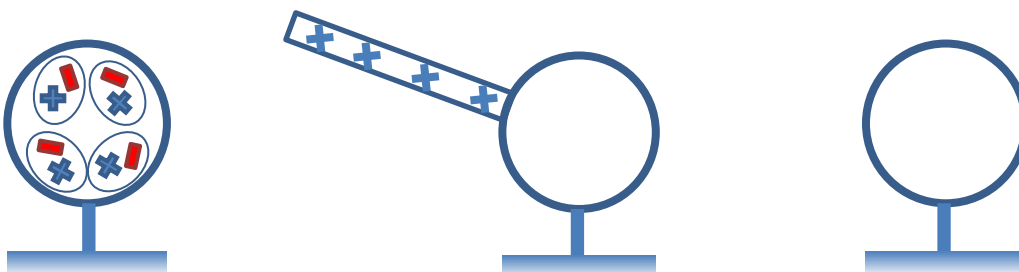
- In CONDUCTORS:



- In INSULATORS:



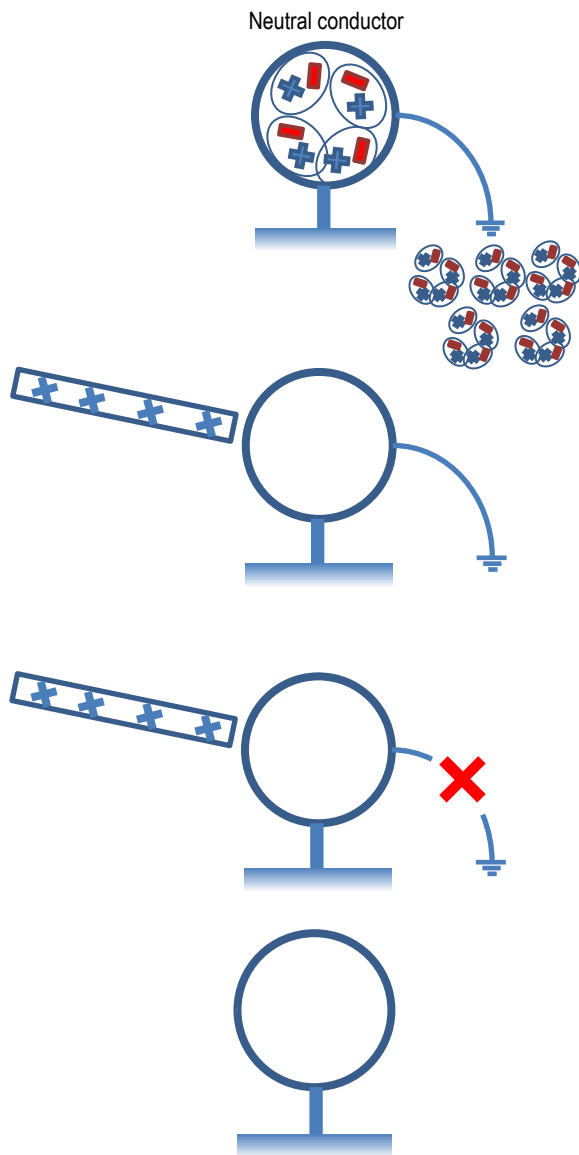
- CONDUCTION = transfer of charges through physical contact → NET CHARGE



CONCEPT: CHARGING BY INDUCTION

- INDUCTION charges object *WITHOUT* touching, CONDUCTION charges object *WITH* touching.

- Object is left with a NET CHARGE.

**Steps for CHARGING BY INDUCTION**

1. Connect neutral conductor to ground
 - Ground = _____ and _____ of charges
 - Sources (**receive | give**), sinks (**receive | give**)
2. Bring + charged rod near conductor → pulls charges:
 - [INTO / FROM] ground
 - [INTO / FROM] conductor
 - If rod was — charged, the opposite would happen.
3. Cut connection between ground and conductor
 - _____ escape of charges!
4. Remove charged rod
 - Conductor is now _____

**CONCEPT: CONSERVATION OF CHARGE**

- Charge [**CAN** | **CANNOT**] be created or destroyed → Known as “charge _____”

- Charge can only be MOVED from one object to another
- This means if one object gains 1 C, the other object loses 1 C

- When conductors are brought together, charges move until they reach _____. → $Q_A = Q_B$

EXAMPLE: In the following scenarios, each pair of conducting spheres is brought into contact and allowed to reach equilibrium. What is the amount of charge transferred, and the direction of transfer, in each of the cases?

-1 C



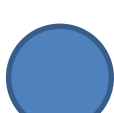
3 C



A

$Q_{\text{Total}} =$	_____
$Q_{\text{Equil}} =$	_____

-3 C



-5 C



B

$Q_{\text{Total}} =$	_____
$Q_{\text{Equil}} =$	_____

3 C



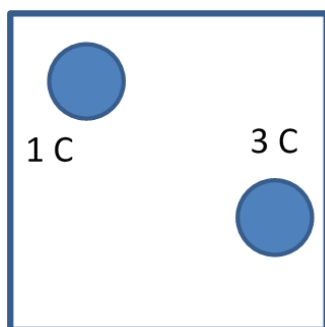
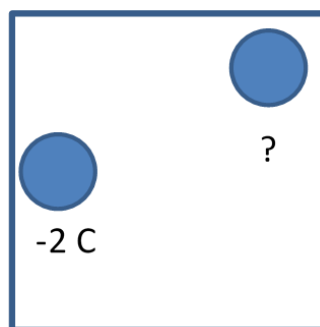
-2 C



C

$Q_{\text{Total}} =$	_____
$Q_{\text{Equil}} =$	_____

EXAMPLE: Two charged, metal balls move around an insulated box, colliding and randomly exchanging charge, but not necessarily reaching equilibrium. Initially, one ball has a charge of 1C while the other has a charge of 3C. After some time, you find that one ball has a charge of -2C. What is the charge of the other ball at this time?

BEFORE**AFTER**

CONCEPT: COULOMB'S LAW

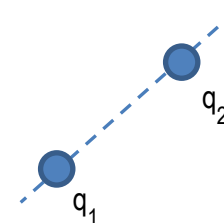
- Electric forces can be _____ or _____.
- Consequence of UNLIKE (+ - / - +) and LIKE (++ / - -) charges

- COULOMB'S LAW gives the force between charges:



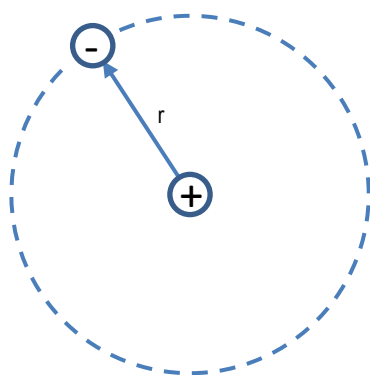
- $F = \underline{\hspace{2cm}}$
- $k = \underline{\hspace{2cm}}$ (Coulomb's constant)
- Units: $\underline{\hspace{2cm}}$

- Force always points along _____
- Like charges [ATTRACT / REPEL], unlike charges [ATTRACT / REPEL]



- **PRO-TIP:** Always find magnitude of Coulomb force by using + numbers → find direction using attract/repel rules.

EXAMPLE: What is the ratio of the electric to the gravitational forces in a hydrogen atom?



Hydrogen Atom	
$G = 6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$	
$M_{\text{Electron}} = 9.11 \times 10^{-31}$	
$M_{\text{Proton}} = 1.67 \times 10^{-27}$	
$r_{\text{prot-elec}} = 5.3 \times 10^{-11} \text{ m}$	

EXAMPLE: If two identical charges are connected by a 5 cm wire with a 10 N tension, what is magnitude of the charges?

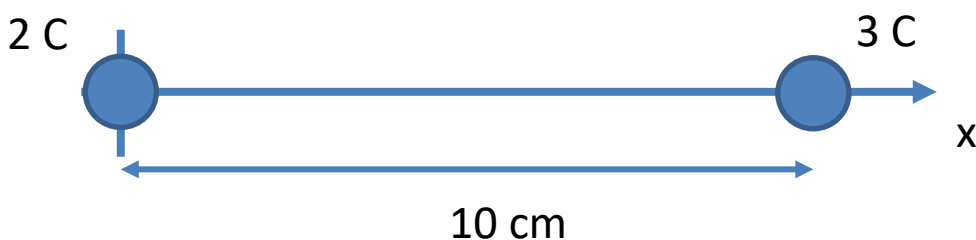


**PRACTICE: CHANGING DISTANCE**

If the force between two charges is F when the distance is d , what will the force between the two charges be if they were moved to a distance of $2d$?

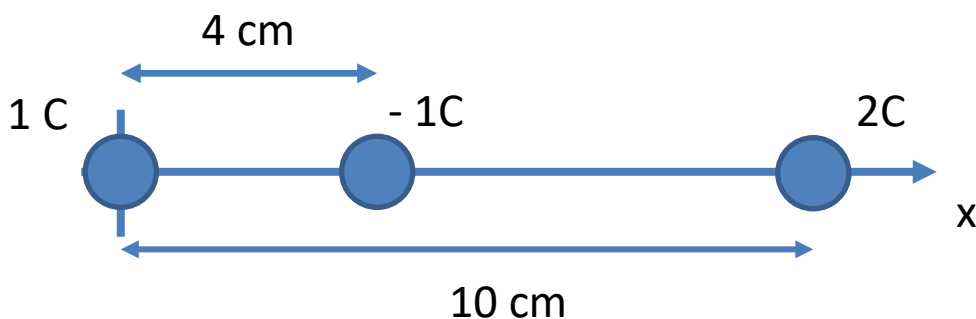
EXAMPLE: CHARGES IN A LINE

Where should we put a 1C charge so that the force on it is zero?

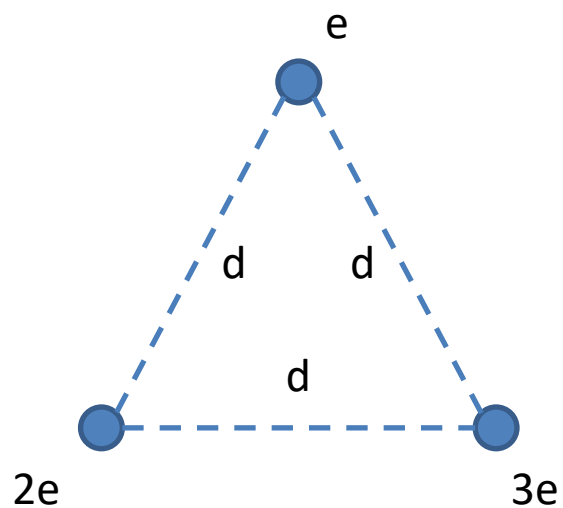


**PRACTICE: 3 CHARGES IN A LINE**

In which direction will the -1 C charge move? If it has a mass of 10 g , what will its initial acceleration be?

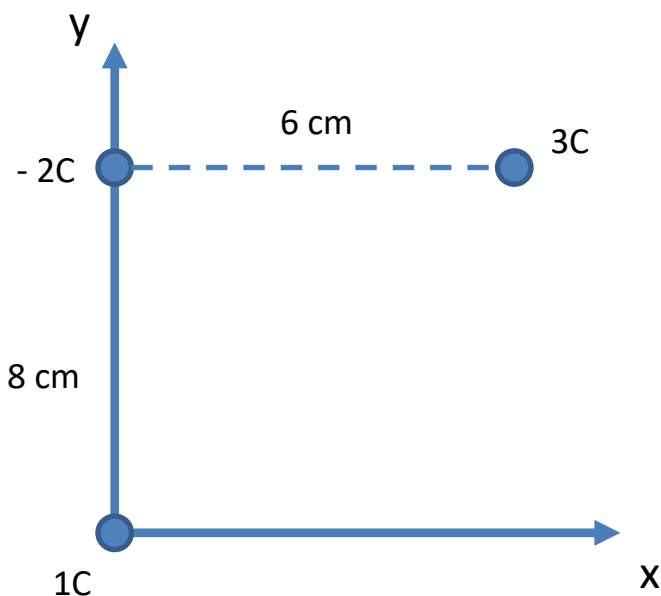
**EXAMPLE: CHARGES IN A TRIANGLE**

Rank all of the possible pairs of charges in the following figure by which pair has the greatest electric force.

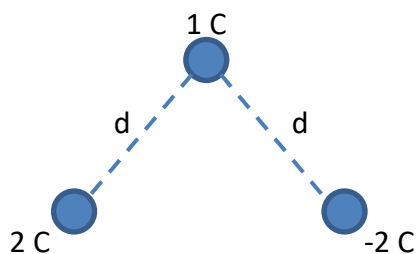
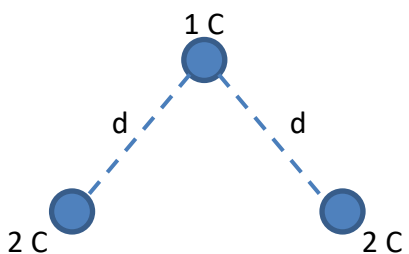


EXAMPLE: CHARGES IN A PLANE

Find the net force on the 3 C charge in the following figure.

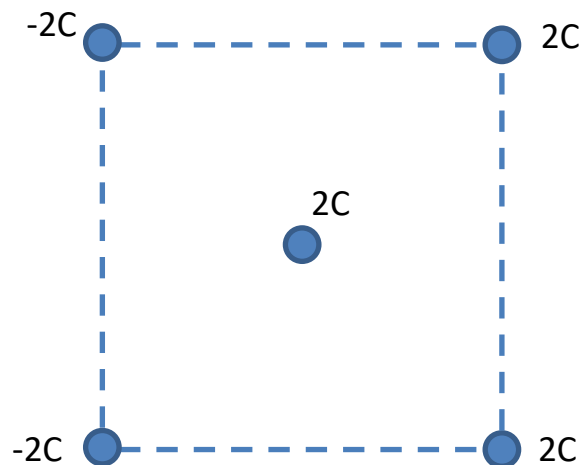
**EXAMPLE: EXPLOITING SYMMETRY IN ELECTRIC FORCES**

For each of the following, what is the direction of the net force on the 1 C charge:

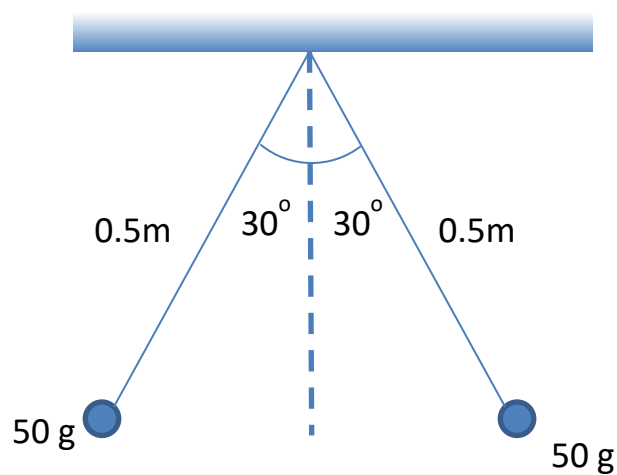


**PRACTICE: DIRECTION OF NET FORCE**

What is the direction of the net force on the charge at the center of the square in the following figure?

**EXAMPLE: ELECTROSCOPE**

Two identical charges at the end of an electroscope's leaves each have a mass of 50 g. If the electroscope leaves are deflected by 30° as shown in the figure, what is the charge at the end of each leaf?

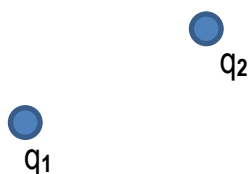


CONCEPT: COULOMB'S LAW WITH CALCULUS

- Coulomb's Law for POINT charges can be given in VECTOR form: $\rightarrow \vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$
 - Unit vector $\hat{r}$ simply gives *direction* between charges
- To simplify, we will use $\vec{r} = r \hat{r}$ $\rightarrow \vec{F} = \underline{\hspace{2cm}}$
 - Vector $\vec{r}$ points along line of charges, like $\hat{r}$

Point Charges

$$\vec{F} = k \frac{q_1 q_2}{r^3} \vec{r}$$



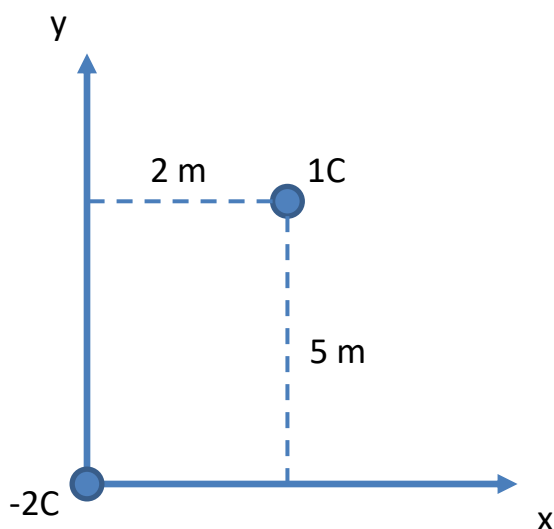
Charge Distribution

- Must tiny pieces of Electric Force

$$\vec{F} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

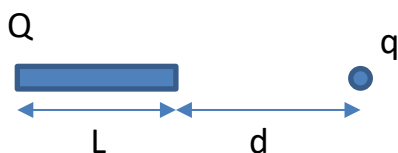


EXAMPLE: What is the electric force on the 1C charge in vector form?



**EXAMPLE: FORCE BETWEEN LINE OF CHARGE AND POINT CHARGE**

What is the force between the line of charge, with a charge Q and a length L , and a point charge q , separated by a distance d as indicated in the following figure? (The linear charge density is $\lambda = Q/L$)

**STEPS FOR COULOMB'S LAW W/ CALC**

- 1) Use geometry to figure out $d\mathbf{F}$ and $\mathbf{r}$
- 2) Relate dQ to integration variable using **charge density**
(λ, σ, ρ)
- 3) Determine limits of integration
- 4) Use integration techniques to solve $\int d\mathbf{F}$

**CONCEPT: ELECTRIC FIELD**

- A single charge will *produce* an **electric field** in all directions that any second charge can *feel* at some distance r .



- Charge q feels a force in an ELECTRIC FIELD, E (set up by Q):

- $F = \underline{\hspace{2cm}}$ $\rightarrow$ Units = $\underline{\hspace{2cm}}$

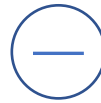
EXAMPLE: A 2C and a 3C charge are separated by some distance r such that the electric field at the 2C charge is 10 N/C. What is the force on the 2C charge?

PRACTICE: BALANCING GRAVITY

A $1.5\mu\text{C}$ charge, with a mass of 50g, is in the presence of an electric field that perfectly balances its gravity. What magnitude does the electric field need to be, and in what direction does it need to point?

**CONCEPT: ELECTRIC FIELD DUE TO POINT CHARGES**

- Positive charges produce fields [**OUTWARD** | **INWARD**], negative charges produce fields [**OUTWARD** | **INWARD**].



- A single charge Q will produce an ELECTRIC FIELD E of magnitude:

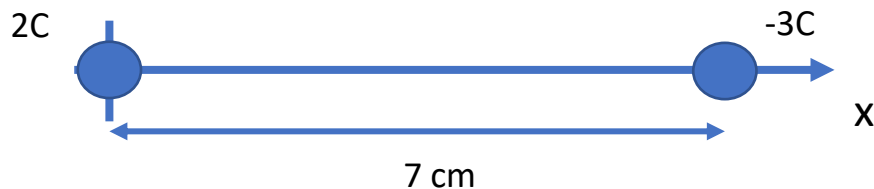
$$E = \underline{\hspace{2cm}}$$

- $r \rightarrow$ distance to **point of interest**. "What is the E -field at ___?"

- A second charge q feels the force F from the Electric Field $E \rightarrow F = qE = \underline{\hspace{2cm}}$.

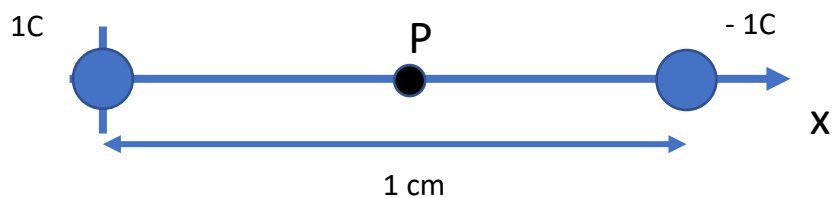
EXAMPLE: At a distance x from a charge Q , the electric field is 20 N/C. At a distance y , the electric field is 10 N/C. What is the ratio, x/y ? Assume x and y lie on the same axis from the electric field source.

EXAMPLE: Two charges lie on the x -axis as shown below. At what point on the x -axis is the electric field zero?

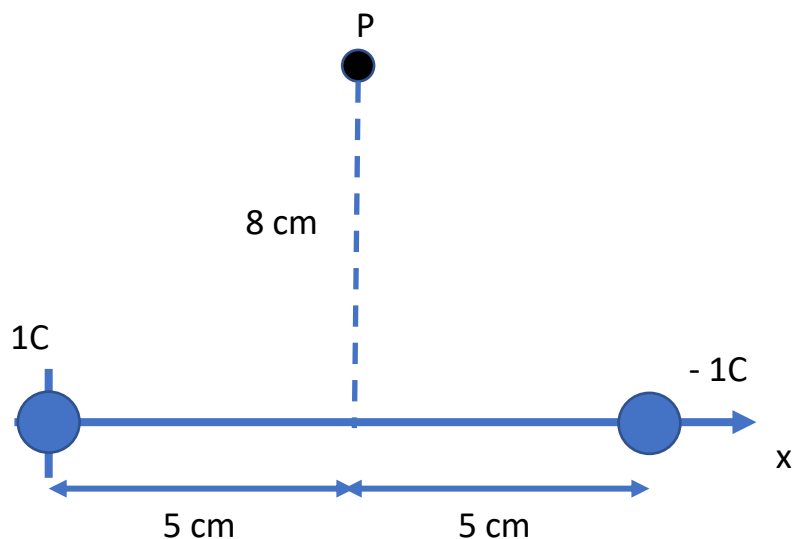


PRACTICE: ELECTRIC DIPOLE

If two equal charges are separated by some distance, they form an electric dipole. Find the electric field at the center of an electric dipole, given by the point P in the following figure, formed by a 1C and a -1C charge separated by 1 cm .

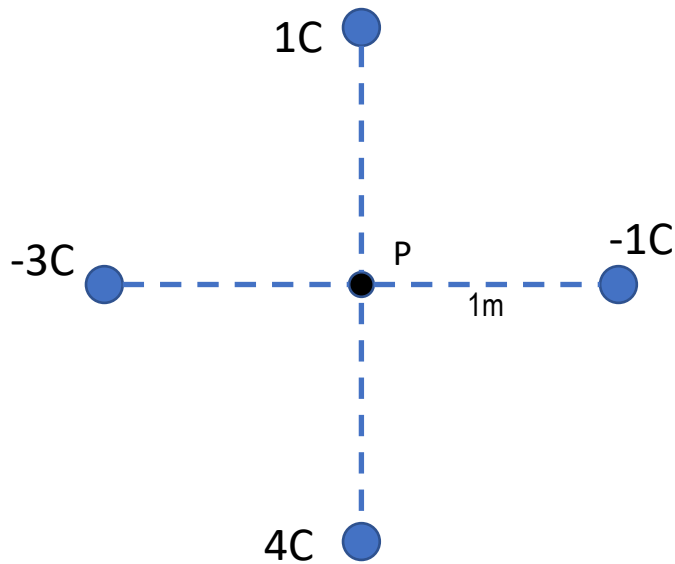
**EXAMPLE: ELECTRIC FIELD ABOVE 2 CHARGES**

What is the electric field at the point above the two charges, indicated as P in the following figure?

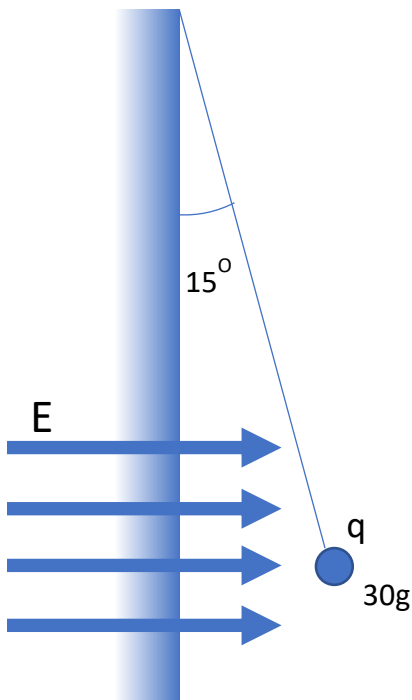


PRACTICE: FIELD AT THE CENTER OF 4 CHARGES

4 charges are arranged as shown in the following figure. Find the electric field at the center of the arrangement, indicated by the point P .

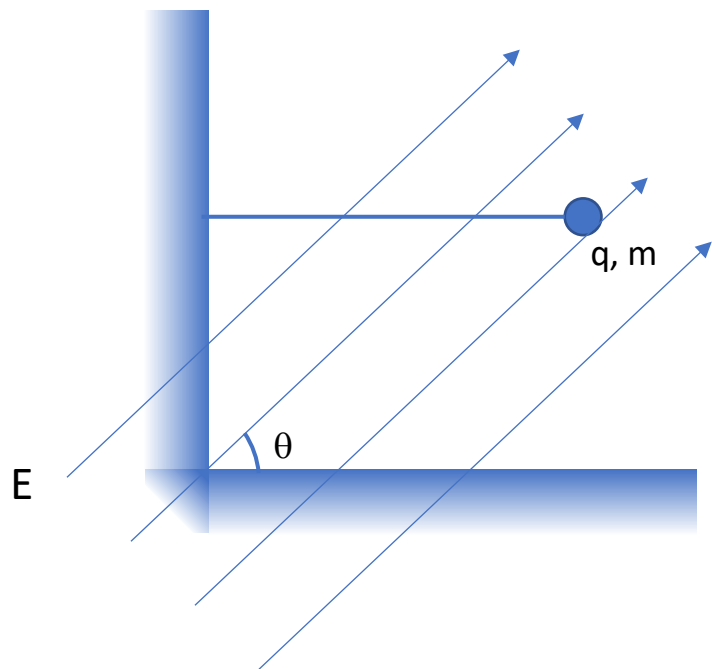
**EXAMPLE: PENDULUM IN ELECTRIC FIELD**

A pendulum is at equilibrium in a uniform electric field as shown in the following figure. If the electric field magnitude is $100N/C$, what is the charge on the end of the pendulum, q ?



**PRACTICE: BALANCING MASS IN UNIFORM ELECTRIC FIELD**

In the following figure, a mass m with charge q is balanced such that its tether is perfectly horizontal. If the mass is m and the angle of the electric field is θ , what is the magnitude of the electric field, E , expressed in terms of m , q , and θ ?



CONCEPT: ELECTRIC FIELDS WITH CALCULUS

- Electric field due to POINT charges can be given in vector form $\rightarrow \vec{E} = k \frac{Q}{r^2} \hat{r}$

- Unit vector $\hat{r}$ simply gives *direction* of $\vec{E}$.

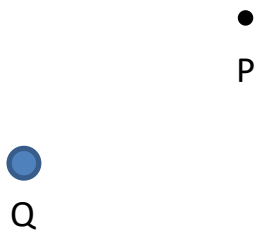
- To simplify, we will use $\vec{r} = r \hat{r}$

$$\rightarrow \vec{E} = \underline{\hspace{2cm}}$$

- Vector $\vec{r}$ points *from* charge **Q** to measuring point **P**.

Point Charge

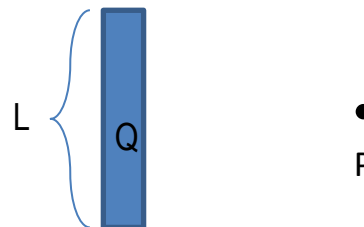
$$\vec{E} = k \frac{Q}{r^3} \vec{r}$$



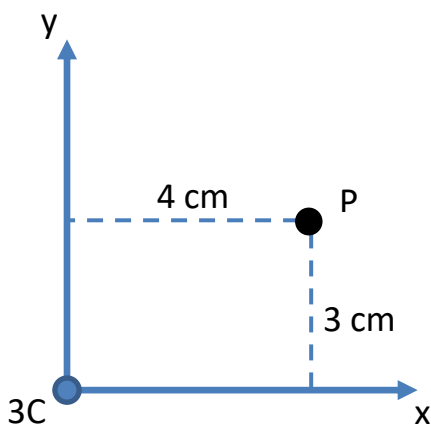
Charge Distribution

- Must tiny pieces of Electric Field

$$\vec{E} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

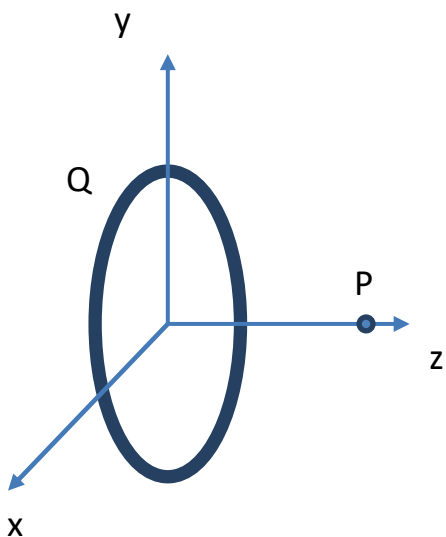


EXAMPLE: What is the vector form of the electric field at the point P indicated in the figure below?



**EXAMPLE: ELECTRIC FIELD DUE TO RING OF CHARGE**

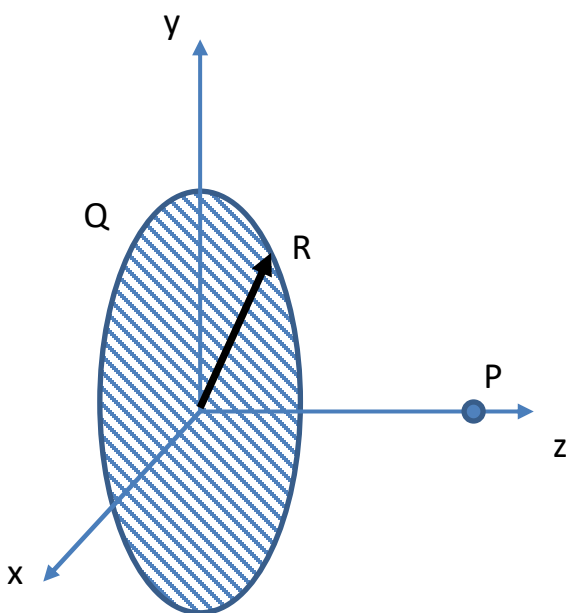
What is the electric field along at point P in the following figure, a distance z from the center of a ring with radius R ?



**EXAMPLE: ELECTRIC FIELD DUE TO A DISK OF CHARGE**

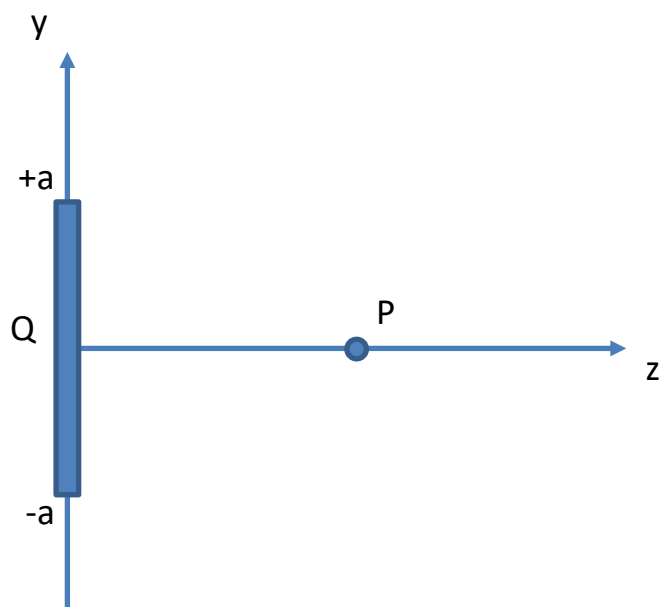
What is the electric field of the following solid disk, with a charge Q , at a distance z along the z -axis, as indicated by P ?

(Hint: The surface charge density is $\sigma = Q/A$)



**EXAMPLE: ELECTRIC FIELD DUE TO LINE OF CHARGE**

What is the electric field at a distance z along the z -axis, as indicated by P, due to a line of charge $2a$ long, with a charge Q , as shown in the following figure?



**CONCEPT: CAPACITORS**

- Two parallel plates of equal and opposite charge produce a _____ electric field between them.
 - UNIFORM = same magnitude everywhere always.
 - Electric field points (**FROM** | **TO**) positive plate (**FROM** | **TO**) negative plate.



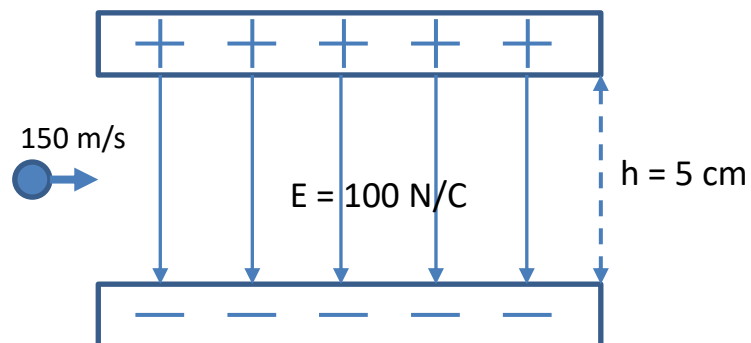
- These are known as CAPACITORS → Things that _____.
- Between plates, $E =$ _____ - Outside of plates, $E =$ _____
- ϵ_0 is the VACUUM PERMITTIVITY, and $\epsilon_0 =$ _____
- Q is the charge on each plate, A is the area of each plate.

EXAMPLE: The electric field between two parallel plates is 1000 N/C. If the plates have an area of 5 cm², what is the charge on each plate?

EXAMPLE: A capacitor produces an electric field E when it is formed by two plates that have charges Q and $-Q$. What happens to the electric field if the charge doubles, but the area of the plates half?

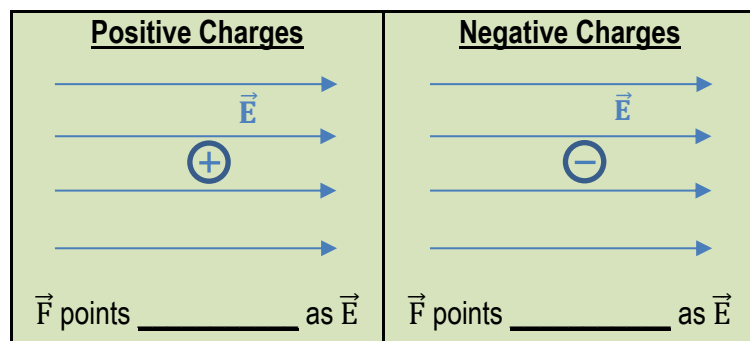
**PRACTICE: KINEMATICS IN A CAPACITOR**

An electron moves into a capacitor at an initial speed of 150 m/s. If the electron enters exactly halfway between the plates, how far will the electron move horizontally before it strikes one of the plates? Which plate will it strike?

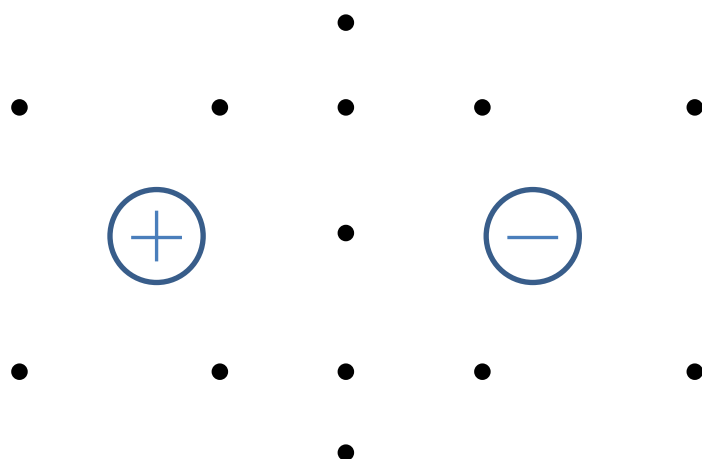


CONCEPT: ELECTRIC FIELD LINES

- Recall that electric field lines point [**FROM** | **TO**] positive charges and [**FROM** | **TO**] negative charges.
- Electric Field lines give the direction a *positive* charge would go:



EXAMPLE: a) Draw the field lines for an electric dipole. b) In what direction would an electron directly between this electric dipole be accelerated?





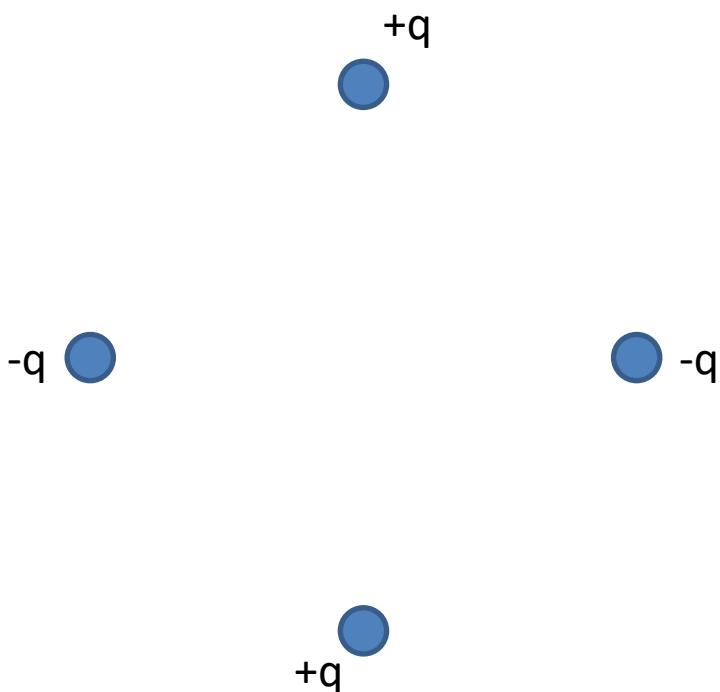
PRACTICE: FIELD LINES OF TWO IDENTICAL, POSITIVE CHARGES

Draw the field lines for a pair of identical, positive charges.



**EXAMPLE: FIELD LINES OF ELECTRIC QUADRUPOLE**

Draw the electric field lines for the four charges shown below. This arrangement is known as an electric quadrupole.



**CONCEPT: INTRO TO DIPOLE MOMENTS**

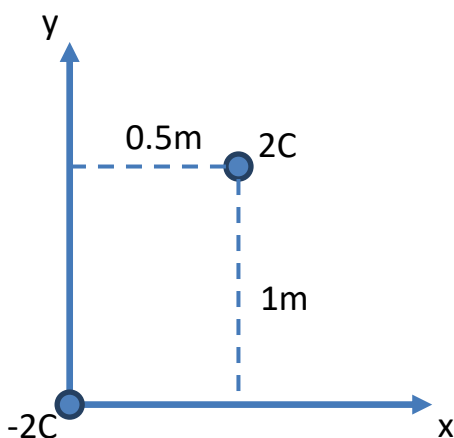
- Two equal charges q with opposite signs (+ and -) form an electric dipole.

- Dipole moment is a vector →

$$\vec{p} = q \vec{d}$$

- $\vec{d}$ is a vector that points [FROM | TO] positive charge [FROM | TO] negative charge.

EXAMPLE: What is the vector dipole moment of the following dipole?



**CONCEPT: ENERGY AND TORQUE OF DIPOLE MOMENTS**

- A dipole in an electric field has potential energy

→

$$U = -\vec{p} \cdot \vec{E} = -pE\cos\theta$$

- A dipole experiences a torque due to an electric field

→

$$\vec{\tau} = \vec{p} \times \vec{E} = pE\sin\theta$$

EXAMPLE: The dipole depicted in the figure below is in a uniform electric field of 200 N/C. What is the potential energy of the dipole? What is the magnitude of the torque the dipole experiences?

